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Task Allocation: Design and implementation of an AI chat assistant in my Streamlit app, including UI/UX changes, data model configuration, and integration with OpenAI

Plan: The plan was to enhance my app by adding an AI chatbot for patient appointment management and to update the UI/UX to feel more modern and professional. Some functions added include:

- AI Chatbot
- Modern UI Update
- Advanced Scheduling

Origin Prompts:

- "Can you build an AI Chatbot for my Streamlit website using the pdf I uploaded as a guideline? I will set the .env folder and add my OpenAI key."
- "Can you install the dependencies and run the app?"
- "Can you restart the app and run it?"
- "Fix this please: AttributeError: module 'openai' has no attribute 'error'"
- "Add in a calendar view UI"
- "Can you also add in advanced scheduling so I can set recurring appointment times."
- "Fix this please: KeyError: 'user_message'"
- "Okay thank you. Can we begin to work on the design of the app?"
- "The core purpose is a patient appointment tracker with an AI assistant. The main user flows are awesome and can remain the same. I want to put a more modern/professional look onto the UI layout and tabs and go for a better color scheme."
- "That looks great."
- "Does this app include all of these things?: login, registration, logout, session state, sidebar, role-based routing, use of multi-page design, at least two distinct user roles, CRUD functionality, JSON data storage, user-specific or role-specific dashboards, clear navigation, usable design, functions and methods, object oriented refactoring, AI assistant connected to OpenAI."
- "Can you connect and push everything to this github repository:
<https://github.com/ryanjdiller/finalproject.git>"

Structural and Feature/UI Analysis/Plans:

Structural Analysis: The app is structured with a sidebar navigation system to differentiate between doctor and patient roles. Features include dynamic session states, role-based routing, CRUD functionalities for appointment data, and the integration of JSON for data storage.

Feature/UI Plan:

- **User Dashboard:** Two distinct user dashboards for patients and doctors, each with access to their specific appointment schedule, chat interface, and a clear visual representation of data.
- **Advanced Scheduling:** The ability to manage and display weekly doctor schedules, support recurring appointments, and prevent double-booking with a calendar integration.
- **AI Chatbot:** Integration of OpenAI for AI responses related to appointment queries and calendar management.

Dated Plan Versions:

- **Version 1.0 (2026-05-12):** Initial setup of AI chat assistant and patient tracking features. Enhanced scheduling features, implementation of recurring appointment functionality.
- **Version 2.0 (2026-05-13):** Added modern features to the app and UI optimization.
- **Version 3.0 (2026-05-14):** Integration of error handling for OpenAI API and calendar-based scheduling for both patients and doctors.

Implementation and Refinements:

- **Step 1:** Integrated OpenAI for the AI Chatbot, allowing for real-time responses based on user queries
- **Step 2:** Added an appointment calendar feature that allows patients to book and view available times
- **Step 3:** Implemented advanced scheduling features for doctors, including recurring appointments and a weekly view calendar
- **Step 4:** Applied error handling in the AI chatbot to prevent crashes
- **Step 5:** Updated the UI with a new modern color scheme and layout to enhance usability